

Early retirement age and the shadow economy in Central European Countries

Bachelor Thesis

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1 Introduction

The term shadow economy is quite common and will not seem new to anyone studying macroeconomics or other economic disciplines. However, despite its seemingly obvious meaning, there is no unified and universally accepted definition of this concept in scientific literature. As rightly noted, «Attempts to measure the shadow economy first face the problem of defining it» (Schneider, Enste, 2000, p. 2), which highlights the fundamental difficulties of research, as well as the importance and methodological sensitivity of choosing the definition itself.

While it is difficult to define the exact date of the emergence of the term shadow economy as an independent concept, it can be said with certainty that since the late 1970s, research interest in the shadow economy, or, as it was then called, «the subterranean economy» (Gutmann, 1977), has grown and expanded significantly, and the research itself has become more in-depth. To date, there is a vast amount of research devoted to this topic, its study and methods of measurement.

One of the most recent and relevant definitions for today is as follows: «The shadow economy (also known as the non-observed, unreported or unregistered economy) comprises various kinds of unreported economic activity (GDP) of registered and unregistered businesses» (EY, 2025, p. 19). In this thesis, we will rely on this definition as the main one.

As for the analysis of the causes and factors influencing the shadow economy, they are often quite narrow and do not take into account the results and conclusions of other social sciences. Therefore, a comprehensive review and scientific analysis of this complex phenomenon is necessary (Schneider, Enste, 2013). As it mentioned earlier, there are quite a few different analyses and approaches to measuring it. However, none of them actually describe or take into account the impact of early retirement. The evidence provided so far is mixed and conflicting. This study was created and written specifically to throw light on this issue.

The relevance of this study is based primarily on the still high global indicators of the shadow economy. Despite the trend towards its gradual decline, according to the latest data for 2023, the volume of unaccounted economic activity (including 131 countries) amounted to 11.8% of total global GDP, which is equivalent to approximately 12.5 trillion US dollars. At the same time, the arithmetic mean for countries reaches 19.3%, which is explained by significant differences in the level of economic development of the countries under study. (EY, 2025)

This thesis focuses on seven Central European countries: Austria, the Czech Republic, Germany, Hungary, Poland, Slovakia, and Slovenia. These countries form a relatively homogeneous group in terms of geographical location, demographic structure, and level of economic development, and they also have similar features in the organisation of their state systems, especially in terms of taxation and pension provision. In addition, they share certain historical similarities. Austria and Germany are developed market economies with stable tax and pension systems that have been formed over decades. The other five countries (the Czech Republic, Hungary, Poland, Slovakia and Slovenia) experienced a similar transition from a planned

to a market economy in the 1990s, accompanied by social system reforms, rising unemployment and an increase in informal employment. Today, all of the countries listed are members of the European Union, which makes them easy to compare: they have similar rules and uniform statistical standards.



Fig. 1 Central European countries examined in this thesis (Austria, the Czech Republic, Germany, Hungary, Poland, Slovakia, and Slovenia)

In view of the above, this study contributes in three ways.

1. It examines the impact of early retirement, an aspect that is almost entirely absent from the existing literature.
2. It examines the tax losses arising from activities within the shadow economy in selected countries.
3. It also assesses the overall level of the shadow economy within the given sample.

The structure of the thesis is as follows. The first part provides background information and introduces the overall idea behind the topic. The second part describes the main objective of the research. The third part presents a literature review and focuses on key factors that influence the shadow economy. The fourth part describes the materials used and the methods applied in the analysis. The fifth part interprets the results and discusses them in the context of the research questions. Lastly, the sixth part concludes the thesis by summarising the key findings and highlighting possible implications.

2 Objectives

The main objective of this study is to examine the possible impact of early retirement on the size of the shadow economy in Central European countries: Austria, the Czech Republic, Germany, Hungary, Poland, Slovakia, and Slovenia. Whilst the shadow economy itself has been widely studied, this specific relationship remains under-researched and requires particular analysis.

To achieve this goal, the thesis sets out three specific objectives:

- To identify whether there is a relation between the level of early retirement and the size of the shadow economy in the countries mentioned.
- Assess the tax losses that may arise as a result of citizens transitioning to informal employment after early retirement.
- Measure the overall level of the shadow economy in selected Central European countries based on available data for 2010–2024.

3 Literature Review

This chapter provides an overview of existing research on the shadow economy. Its main aim is to present a comprehensive review of the topic and identify factors that directly or indirectly influence its size. It also discusses the main approaches used to study, measure and analyze the underground economy. Particular attention is paid to gaps that require further study and elaboration.

This section gives particular attention to the following groups of factors:

- **Economic factors** — characteristics describing the overall state and structure of the economy, such as GDP per capita, unemployment, and self-employment.
- **Fiscal and tax factors** — elements related to the tax system and state finances, including corporate income tax, personal income tax, social security contributions, and government effectiveness.
- **Socio-demographic factors** — characteristics of the population that influence economic behaviour, primarily early retirement and migration.

Considering these factors provides a clearer understanding of how various characteristics of the economy and society affect the level of the shadow economy. It also helps to identify differences between Central European countries and provides a basis for analysing the possible impact of early retirement on the size of the shadow economy.

This chapter is structured as follows. At first, the general drivers of the shadow economy are examined. Then, individual factors are analysed based on the results of previous studies and a comparison of their impact in Central European countries, both individually and in the overall picture. This approach helps to form a theoretical basis for the subsequent analysis and interpretation of the research results.

3.1 Factors Causing Shadow Economies

3.1.1 Definition and Theoretical Background

In order to move on to a qualitative explanation of the causes and an overview of the factors influencing the size of the shadow economy, it is first necessary to clarify what should and should not be considered part of the shadow economy.

It is worth mentioning that the shadow economy has many names, such as hidden economy, grey economy, black economy, cash economy, or informal economy. All these terms describe different forms of shadow economic activity (Medina, Schneider, 2018). In other significant work, note that there are about 40 adjectives and 19 nouns used to describe this phenomenon in scientific literature, not to mention the number of possible combinations (Koufopoulou et al., 2019). However, it is important to understand that even while using the same terms or describing the same phenomenon, authors may mean completely different things.

Tab. 1 Revised adjectives and nouns used to describe the shadow economy

Adjectives			
Bazaar	Black	Cash – in – hand	Clandestine
Dual	Envelope	Falsely – declared	Grey
Hard – to tax	Hidden ¹¹	Illegal ¹²	Illicit
Informal	Invisible	Irregular	Moonlight
Non – Observed	Occult	Off – the – books	Parallel
Peripheral	Precarious	Second	Secret
Shadow	Submerged	Subterranean	Twilight
Undeclared	Underground	Unlawful	Unlicensed
Unobserved	Unofficial	Unorganized	Under – reporting
Under – taxed	Unrecorded	Unregulated	Unreported
Nouns			
Activity	Banking	Cash	Economic activity
Economy	Employment	Enterprise	Entrepreneurship
Exchange	Firm	Income	Labour
Market	Salaries	Sphere	Sector
Trade	Wage	Work	

Source: Koufopoulou et al., 2019

For example, Vito Tanzi writes in his work: “The shadow economy is often confused with the informal economy that has other roots and causes” (2003, p. 2). At the same time, the International Monetary Fund gives the following definition: “The informal economy, comprising activities that have market value and would add to tax revenue and GDP if they were recorded” (IMF, 2020). This illustrates the conceptual confusion surrounding the term.

When using the term informal economy, Vito Tanzi refers to a more “natural” or “traditional” informal economy, non-market activities that existed even before the formation of modern states and tax systems. Meanwhile, in the IMF definition, the term informal economy is actually used as a synonym for shadow economy. That is why every research paper on this topic must specify what exactly the definition covers and what types of activities it refers to.

Moving on to a more specific question — what exactly does the concept of shadow economy encompass?

The classification presented in the System of National Accounts (SNA, 1993) model, on which the OECD approach is also based, divides the shadow economy into four sectors:

- Underground production – legal and productive activities that are deliberately concealed from public authorities to avoid paying taxes or following official regulations.
- Illegal production – production or trade of goods and services that are forbidden by law or conducted by unauthorised producers.
- Informal sector production – productive activities carried out by small, unregistered enterprises or self-employed individuals, usually within the household sector.

- Production of households for own final use – productive activities that create goods or services consumed directly by the households that produced them (OECD, 2002, pp. 13–14).

Later, some researchers developed this system further, adding another category:

- Statistical underground – this group includes legal and productive activities that should be recorded in official statistics but are missed because of deficiencies in data collection systems (Koufopoulou et al., 2019).

In contrast, Bagachwa and Naho propose a more conceptual division of the so-called «second economy», dividing it into three main sectors:

- Informal sector – very small, mostly unregistered economic activities by individuals or households, both in urban and rural areas.

- Parallel market activities – production or trade of legal goods and services that are sold illegally, outside official markets.

- Black market activities – trade and production of goods or services that are strictly prohibited by law. (1995)

Thus, both classifications overlap in many aspects, distinguishing between illegal activities, hidden legal activities, and the informal sector. However, their emphases differ. The OECD/SNA approach focuses primarily on the statistical accounting of all economic activity, including that which does not pass through the market but contributes to the country's aggregate production. Therefore, this system distinguishes the category of production for its own final use, as it affects GDP calculations even if it is not related to sales. In contrast, the Bagachwa and Naho model focuses on market forms of activity, i.e., activities that generate income or participate in exchange. Production for own consumption is not included in their classification, as it is not market-oriented and does not reflect economic behaviour aimed at making a profit.

The addition of the Statistical Underground category emphasises that not all gaps in statistics are caused by deliberate concealment of activity. Some economic activity remains outside official records due to limitations and imperfections in statistical systems.

Thus, in its broadest sense, the phenomenon of the shadow economy encompasses both legal and illegal activities, both deliberately concealed and unaccounted for technical reasons, as well as market and non-market forms of economic activity.

At the same time, Tanzi, mentioned earlier, in conjunction with the approach of Bagachwa and Naho, considers the shadow economy exclusively as market activity: «It is wise to keep the informal economy and the shadow economy as two separate phenomena, even though they intersect significantly» (Tanzi, 2003, p. 2).

Some fundamental works, on the contrary, include only the legal part of market activity in the concept of the shadow economy. For example, Schneider, Buehn, Montenegro (2010) define it as follows: «The shadow economy includes all market-based legal production of goods and services that are deliberately concealed from public authorities to avoid payment of income, value-added, or other taxes».

In general, considering all of the above, it can be said that there is no universal and completely unambiguous definition of the shadow economy. It is interpreted differently in various sources — as a combination of illegal, informal or simply unaccounted activities. However, for the purposes of this study and to ensure its relevance, the shadow economy is considered to be the aggregate of all forms of unobserved economic activity, covering both registered and unregistered entities whose activities are not reflected in official reports, but which have an impact on the total gross domestic product. (EY, 2025)

3.1.2 General Causes of the Shadow Economy

Certainly, the simplest and most obvious reason for the existence of the shadow economy is personal gain. "Individuals tend to participate in the shadow economy when the perceived benefits outweigh the risks and costs associated with potential sanctions" (Czapkiewicz et al., 2022, p. 271). In this context, individuals really act as "rational calculators" who choose the most profitable side of economic activity for themselves (Medina, Schneider, 2021). This is entirely consistent with the basic principles of economic theory, according to which, the main motive for the actions of economic actors is the desire to maximise profits.

As in the case of the definition of the shadow economy, there are many approaches to classifying the factors that influence its scale.

One of them suggests dividing the factors into the following three categories:

- Longue-durée factors — factors that have developed historically and are passed down from generation to generation. They are related to the culture, traditions, mentality and historical experience of the state.
- Institutional and regulatory factors — these are current, changeable political and economic conditions that determine the "rules of the game" and influence how profitable it is for people to work officially.
- Micro-level and perception factors — individual motivations and people's perceptions of the system, including psychological and behavioural aspects (Myck, Tyrowicz, Wojcik, 2015).

Another relevant model considers taxpayer behaviour (including the tendency to avoid taxation) as a multi-level system influenced by a combination of five main groups of factors:

- Business factors — describe the internal characteristics of the enterprise or self-employed person: structure, scale, financial stability, form of ownership (sole proprietorship, partnership, company, trust), and level of accounting.
- Industry factors — reflect the specifics of the industry in which the taxpayer operates, including the degree of competition and profitability, seasonality, level of regulation, cost and income structure, and the presence of professional associations or trade unions.
- Sociological factors — related to the social environment and cultural attitudes in which the taxpayer operates: attitude towards the state, trust in institutions, level of education, age, gender, cultural and ethnic characteristics.

- **Economic factors** — cover external economic conditions that determine incentives and risks for doing business: investment levels and interest rates, labour market conditions, the tax system, government policy, and inflation.
- **Psychological factors** — reflect internal attitudes, motivation and perception of the fairness of the system: fear of punishment, sense of justice, trust in tax authorities, personal values (greed, morality, risk appetite). (OECD, 2004)

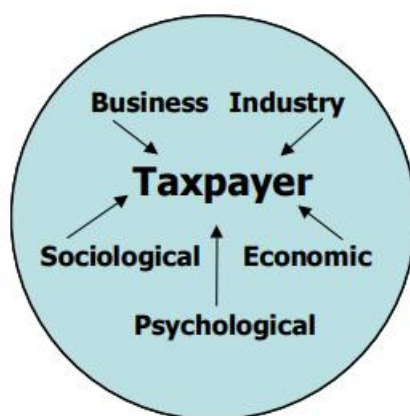


Fig. 2 Factors influencing taxpayer compliance behaviour

Source: OECD, Compliance Risk Management: Managing and Improving Tax Compliance, 2024.

There is also an approach that divides factors into three more categories, related to taxpayer attitudes and perceptions:

- **Direct factors** — related to the work of the tax administration and the tax system itself. These include tax rates, tax structure, audit probability, and quality of administration.
- **Indirect factors** — reflect the stable characteristics of taxpayers themselves, such as demographic and social parameters: age, education, type of employment, form of business, and industry.
- **Taxpayers' attitudes and perceptions** — describe how people perceive the tax system and the state as a whole. These include trust in the authorities, perceptions of fairness, social norms and risk appetite (European Commission, 2023).

Despite the differences in approach, all classifications reflect similar patterns: the influence of economic, institutional, social and psychological factors on taxpayer behaviour. Myck, Tyrowicz, Wojcik (2015) focus on historical and institutional differences, the OECD (2004) model on structural features and the business environment, and the European Commission (2023) on perceptions and trust in the state. Overall, these approaches can be seen as complementary, as they cover different levels of analysis.

It is important to note that with the development of technology and digitalisation, new forms of economic activity are arising that also affect the scale of the shadow economy. The emergence of cryptocurrencies, online platforms and digital

transactions shows that technology is transforming the structure and forms of informal economic activity. However, concepts such as corruption, tax reform, and transparency remain central (Arisanti, Puspitasari, 2024).

And speaking of the most significant factors. According to a recent study identifying the key factors influencing the development of the shadow economy up to 2025, the most significant drivers were reported to be a high tax burden (92%), recession in the local economy (91%), and the complexity of tax systems (91%). Additionally, the potential for a global economic downturn, ease of participation in the informal sector, and ease of tax avoidance were all identified by over 80% of respondents as contributing to the growth of the shadow economy (ACCA, 2017).



Fig. 3 Assessment of economic factors' potential for increasing the scale of the SE

Source: ACCA research, 2017.

Based on the approaches presented and the key drivers identified, this study focuses on the following factors affecting the shadow economy: GDP per capita, unemployment rate, self-employment, corporate income tax, indirect taxes on employees, and social security contributions. These indicators reflect key economic, fiscal, and institutional dimensions.

In addition, the study pays specific attention to factors such as early retirement and migration, as they are closely interrelated. However, the impact of early retirement on the shadow economy remains understudied in existing literature.

Based on the above classifications of factors, this study proposes a version of their grouping adapted for the purposes of this thesis:

- Economic factors: GDP per capita, unemployment rate, and self-employment.

- Fiscal and tax factors: corporate income tax, direct taxes (taxes on employees), social contributions, and government effectiveness.
- Socio-demographic factors: early retirement, migration.

In summary, it can be stated that the shadow economy is formed not by a single factor, but by a combination of them. Historical, institutional, and social conditions create the foundation, while economic drivers and personal motivation determine the extent of participation in informal activities. This balance varies from country to country, making regional comparisons particularly important. That is why the next section of this chapter focuses on the characteristics of the shadow economy in Central European countries.

3.1.3 The Case of Central European Countries

The countries of Central Europe (Austria, the Czech Republic, Germany, Hungary, Poland, Slovakia, and Slovenia) form a region united not only by their common geographical location, but also by their similar historical foundations, religious and cultural roots, and, importantly, their similar levels of economic and demographic development. This area, located between Western and Eastern Europe, can be seen as a kind of bridge between the two regions.

At the beginning of the 20th century, some of these countries were part of the German or Austro-Hungarian empires, and later a significant part of the region came under the influence of the Soviet Union (EBSCO, 2024). Germany was the first to join the European Union in 1957. Austria followed in 1995. The Czech Republic, Poland, Hungary, Slovakia, and Slovenia joined the European Union in 2004, which gave a powerful impetus to the organic economic development of Central European countries as they became part of the single European market (Török, 2025).

It should also be pointed out that the countries selected for this study can be divided into two categories. The first group includes Germany and Austria, which are among the most prosperous countries in the world and have more developed economies. The second group brings together the other countries in the region, which joined the European Union later and have followed a slightly different path of economic development. In this regard, Germany and Austria will be considered as a kind of benchmark and the main factor for comparison in this thesis.

The similarity in the economic development of the countries in the region is largely due to their developed infrastructure, strong industrial sectors, and sustainable agriculture.

In addition, Central European countries share similar demographic trends, such as an ageing population, declining birth rates, and, as a result, a decline in the total population. These countries also face the ongoing problem of migration of the most skilled labour to the West (Török, 2025).

Therefore, the similar economic and demographic conditions of these countries create the base for the shadow economy to grow. In the following, we will look at some stats that show how big it is in Central Europe.

According to the latest studies presented in various sources, the share of the shadow economy in the European Union varies from 6.6% to 21.9% (CzechTrade

2024; Achim et al. 2024; EY 2025). The spread of indicators is explained by the use of different assessment methods and research approaches. Nevertheless, the fact remains that the shadow economy still accounts for a significant part of the GDP of EU countries.

One of the latest analytical works by EY (2025) presents comparative statistics on the level of the shadow economy by country. According to the data provided (see figure below), the lowest indicators are recorded in the Czech Republic (6,7%), Austria(5,7%), and Germany(6,8%) — these countries are at approximately the same level. The highest indicator is observed in Poland and amounts to 9.7%. The variation between countries in the region is approximately 3.2%. This indicates a small range of fluctuations, confirming the structural and economic similarities between Central European countries. To better understand the factors influencing these differences, it is important to consider key economic indicators for the region, such as GDP per capita.

3.2 GDP per capita

«GDP per capita is the most robust predictor of the size of the informal economy. The most straightforward interpretation of the results in this section is that the informal economy is a manifestation of underdevelopment» (La Porta, Shleifer, 2008). Overall, this confirms a consistent relationship: the higher the level of income in a country, the smaller the scale of the shadow economy.

However, new studies provide additional perspective. For example, one analysis shows that the relationship between income and the shadow economy may be non-linear. At low-income levels, GDP growth per capita does indeed reduce the size of the shadow economy, but after a certain level, richer countries sometimes show a new increase in informal activity, which may be linked to flexible forms of employment or voluntary informality in developed economies (Wu, Schneider, 2019). Therefore, recent sources emphasise the need to take such dynamics into account when measuring the role of income in shaping the informal sector.

To assess how this relationship manifests itself in Central European countries, it is important to refer to current statistical data. According to the World Bank methodology, GDP per capita is expressed in current US dollars per person and represents the average level of economic output and income in each country.

Tab. 2 GDP per Capita in Selected Central European Countries, 2024 (current US\$)

Country Name	2024
Austria	56833.19605
Czechia	31706.62342
Germany	55800.21945
Hungary	23310.75127
Slovenia	34089.35277
Slovak Republic	26147.90284
Poland	25022.67164

Source: World Bank, World Development Indicators, 2024.

World Bank data for 2024 reveal a noticeable gap within the Central European region. The highest levels of GDP per capita are observed in Austria and Germany, reflecting their more advanced economic structures and higher productivity. In contrast, GDP per capita in Slovenia, Czechia, Slovakia, Poland, and Hungary remains substantially lower, amounting to roughly half of the levels recorded in Austria and Germany. Differences are also evident within this group, indicating heterogeneity in economical performance among the remaining Central European countries.

Overall, these data indicate that income disparities within the region are significant. This observation is consistent with findings in the literature suggesting a relationship between the level of economic development and the scale of the shadow economy. To complete the picture, it is useful to examine how labour market indicators, particularly the unemployment rate, fit into this dynamic.

3.3 Unemployment

Unemployment is traditionally considered one of the key factors influencing the scale of the shadow economy. During periods of economic recession, when formal employment opportunities become scarcer, people are more inclined to seek alternative sources of income outside the official labour market, which increases the likelihood of them turning to informal activities (World Bank, 2010). This mechanism is particularly evident in economies with weaker institutional controls, where the cost of entering the shadow economy is relatively low.

Empirical studies confirm a consistent link between the level of unemployment and the size of the shadow economy. In general, rising unemployment contributes to the expansion of the informal sector, but the strength of this effect depends significantly on the quality of institutions and the effectiveness of public administration (Elgin, Oyvat, 2019). In countries with stable institutions, the negative impact is mitigated, while in more vulnerable economies, unemployment and shadow activity reinforce each other. This is why analysing unemployment dynamics in Central European countries is important for understanding the differences in the scale of the shadow economy within the region.

According to the latest Eurostat data, the unemployment rate in the European Union remains relatively stable. In September 2025, the EU unemployment rate stood at 6.0%, which is only slightly higher than in September 2024. Overall, unemployment in the European Union increased by 0.2 percentage points, corresponding to approximately 227,000 additional unemployed persons.

Tab. 3 Unemployment Rate in Selected Central European Countries, 2024 (% of total labour force)

Country Name	2024
Austria	5.161
Czechia	2.603
Germany	3.372
Hungary	4.466
Slovenia	3.697
Slovak Republic	5.335
Poland	2.807

Source: World Bank. World Development Indicators, 2024.

However, the situation in Central Europe looks much more favourable. Based on World Bank data for 2024, countries in the region exhibit consistently low unemployment rates compared to the European Union average. Across the selected Central European countries, unemployment generally ranges between approximately 2.6% and 5.3%, indicating a relatively high degree of labour market stability.

Looking at individual countries, the lowest unemployment rates in the region are recorded in Czechia and Poland, while Germany and Slovenia also display comparatively low levels of unemployment. Austria and the Slovak Republic are located at the upper end of the regional range, whereas Hungary records moderately higher unemployment relative to other countries in the sample. Nevertheless, all selected Central European countries continue to demonstrate unemployment rates that remain below the EU average.

This context is also important for analysing the shadow economy, as persistently low unemployment in Central Europe, especially in countries such as Czechia and Germany, reduces incentives to move into informal employment and partly explains why the region exhibits a relatively low level of shadow economic activity.

3.4 Self-employment

Despite the importance of unemployment, it reflects only one aspect of the labour market situation. Self-employment is an equally significant factor closely linked to the scale of the shadow economy. Studies note that the share of self-employed persons in total employment can be considered one of the indicators of the scale of labour activity in the shadow economy (Schneider 2012). Therefore, it makes sense to further analyse the role of self-employment in Central European countries and how it affects the scale of the informal sector generally.

It is also worth mentioning the heterogeneity of this concept. Though self-employment formally means “working for oneself”, this term encompasses a wide range of employment models. In developing economies, self-employment is more often associated with informal activities that arise not out of choice but out of necessity. For example, one of Keith Hart's key works on the shadow economy, using Ghana as an example, shows that official figures reflected high unemployment and low GDP, while in practice a significant part of the population was self-employed and worked in the informal sector. The main reasons for this were poverty, limited access to formal employment, and weak institutions (1973).

In more developed economies, self-employment is already considered a legal form of employment as freelancers, consultants, or individual entrepreneurs who officially conduct their activities. However, problems related to administrative barriers remain. Early studies, such as de Soto's study of Peru, showed that a number of entrepreneurs deliberately avoided registering their businesses due to excessive bureaucracy and the high cost of entering the formal market. (1989)

Furthermore, researche emphasise that self-employment reacts to the economic cycle. This refers not only to “forced” self-employment during periods of recession, but also to procyclical transitions to self-employment during times of economic growth. For example, Taylor (1996) shows that in favourable periods, some workers consciously switch to self-employment, taking advantage of the low risk of being left without formal work in case of failure. Maloney (1998) comes to similar conclusions, demonstrating that in several Latin American countries, the proportion of self-employed workers increases precisely during periods of economic upturn.

Considering the theoretical heterogeneity of self-employment, it is reasonable to turn to statistical evidence and examine how this form of employment is structured in practice across Central European countries.

Tab. 4 Self-Employment as a Share of Total Employment in Selected Central European Countries, 2024 (%)

Country Name	2024
Austria	12.2
Czechia	16
Germany	8.9
Hungary	12.7
Slovenia	14
Slovak Republic	15
Poland	19.6

Source: OECD Statistics, 2024.

According to OECD data for 2024, the share of self-employed individuals differs substantially across the region. The highest levels are observed in Poland, where self-employment accounts for nearly one-fifth of total employment, followed by Slovakia and Slovenia. Austria and Hungary also record relatively high shares of self-

employed workers, while Germany and Czechia exhibit noticeably lower levels of self-employment compared to other countries in the sample.

These differences reflect the diverse institutional, regulatory, and labour market environments across Central Europe. In countries with a higher prevalence of self-employment, this form of labour market participation may include a broad mix of genuinely entrepreneurial activity and more necessity-driven or marginal forms of self-employment. In contrast, lower shares of self-employment may indicate a stronger role in standard employment contracts and higher absorption of labour into the formal wage sector.

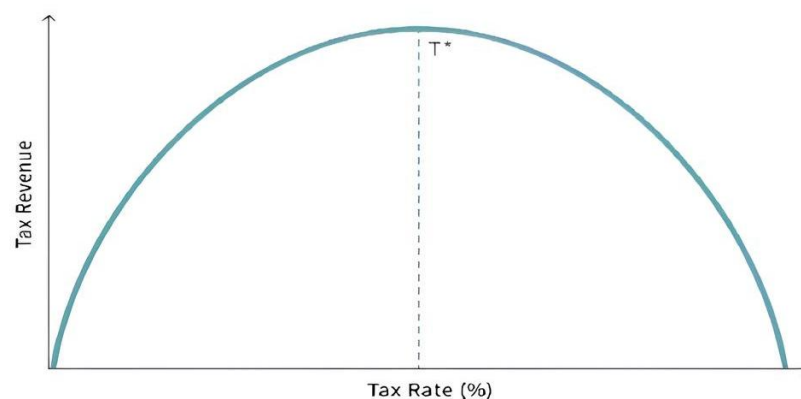
The observed cross-country variation indicates that self-employment constitutes an important but structurally diverse component of labour market structures in Central Europe. Differences in the prevalence of self-employment reflect broader institutional and economic conditions, suggesting that this form of employment interacts with other structural factors rather than independently determining the scale of the shadow economy.

It is therefore logical to move on to consider fiscal and tax-related factors, including corporate income tax, which are more directly linked to incentives for informal economic activity and the overall scale of the shadow economy.

3.5 Corporate income tax

The tax burden affects not only budget revenues, but also the motivation of economic agents to remain in the formal sector. Lower rates reduce incentives for evasion and transition to the gray economy, while tax increases increase pressure on supply and may limit economic activity. This is reflected in the Laffer curve concept, according to which there is a critical tax point T^* , and exceeding this point leads to a decrease in tax revenues. In such a situation, some households and firms reduce their legal activities, move their operations to other countries, or go underground (Fuchs, 2019).

Fig. 4 The Laffer Curve and the Critical Tax Rate



Source: Laffer, 2004.

An additional perspective comes from research showing that the structure of factors influencing the shadow economy varies significantly between countries. In developing economies, weak institutions and a high proportion of informal entrepreneurship are key factors. In developed countries, the main factor is the level of tax burden (EY, 2025). This means that tax rates, including corporate tax, indirect taxes, and social contributions, have a significant impact on the size of the shadow economy in economically stable countries, which include Central Europe.

Considering corporate tax separately, many studies confirm a direct correlation between its rate and the size of the informal sector. One study showed that a 10-percentage-point increase in the effective corporate tax rate leads to an increase in the share of the informal economy by approximately 2 percentage points. This indicates that part of the business goes underground because it becomes too costly to operate legally (Djankov et al., 2010). This conclusion is consistent with the logic of the Laffer curve, which links the level of taxation to the incentive to remain in the formal sector. However, this relationship is more pronounced in developing countries, where the tax burden is higher, and the institutional environment is weaker.

A study based on data from Estonia, Latvia, and Lithuania showed the opposite result compared to most international studies. In these countries, an increase in the effective corporate tax rate was accompanied by a reduction in the shadow economy. The authors explain this by the limited opportunities for capital flights offshore and specific features of national administration. This example shows that there is no universal relationship between the corporate tax rate and the size of the shadow economy, and that the effect depends on the institutional context (Giedraitis et al., 2023).

Given this difference in mechanisms and the sensitivity of the shadow economy to the tax burden, we will now move on to analyse selected Central European countries.

Tab. 5 Corporate income tax in Central Europe (2025)

2025 Corporate Income Tax Rates in Europe Tax Foundation	
Country	Combined Statutory Corporate Income Tax Rate
Hungary (HU)	9.0%
Poland (PL)	19.0%
Czechia (CZ)	21.0%
Slovakia (SK)	21.0%
Slovenia (SI)	22.0%
Austria (AT)	23.0%
Germany (DE)	29.9%

Source: OECD Tax Database, 2025.

Data for Central European countries show that combined statutory corporate income tax rates range from 9% in Hungary to 29.9% in Germany. The average regional range is 20.7%. Hungary has the lowest CIT rate in the European Union, while

Germany has one of the highest. It is noteworthy that the most economically developed countries in the region, Germany and Austria, apply higher rates. This reflects their stable fiscal systems and ability to administer taxes effectively without encouraging shadow activity.

As a result, significant differences in CIT rates do not lead to the expected differences in the scale of the shadow economy. Germany maintains a low level of shadow activity despite its high rate, while Hungary does not demonstrate any noticeable advantages despite its record low CIT. This confirms that the impact of corporate tax is not universal. In developed economies, the quality of institutions, the level of trust and administrative efficiency play no less important role than tax rates themselves.

It is therefore logical to consider indirect taxes, which also influence the behaviour of market participants.

3.6 Personal income tax

The level of personal income tax can significantly affect the size of the shadow economy. As the tax burden increases, so does the motivation to earn part of one's revenue unofficially. However, a direct correlation is not always apparent, as the impact of income tax overlaps with other economic factors. Nevertheless, one study shows that a one percentage point increase in the marginal federal income tax rate in the United States leads to an increase in the shadow economy of approximately 1.4 percentage points (Schneider, Enste, 2002, p. 15). This confirms that higher rates can push workers and firms into the shadow economy. It also corresponds to the Laffer effect and the tax wedge described earlier.

Different groups react differently to income tax increases. Self-employed individuals are more likely to underreport their actual income, as their earnings are more difficult to verify and control. Employers and employees start using cash-in-hand wages to avoid declaring the full amount to the state. According to a survey conducted in Latvia, 67% of respondents believe that people agree to such schemes precisely because of the high tax burden on labour (Myck, Tyrowicz, Wojcik, 2015). In some cases, a person officially receives only the minimum income, and the rest is paid informally. Small firms also seek to reduce their tax liabilities. Most often, this takes the form of partially underreporting revenue, making cash payments without recording transactions, or employing unregistered workers. Such practices allow companies to reduce their tax base and lower their expenses.

Returning to the above-mentioned study, which is partly devoted to the impact of taxes on participation in informal employment, it is important to note another of its findings. According to the survey results, both high-income respondents (69.9%) and low-income respondents (68.1%) agree that people participate in shadow employment mainly because of the high tax burden (Myck, Tyrowicz, Wojcik, 2015). This result confirms that the perception of income tax as too high a burden can create similar motivations for informal activity among different social groups.

These results are also relevant for Central European countries, as their economic structure is to some extent comparable to that of Lithuania. These countries have similar conditions in which income tax influences the decisions of households and workers regarding participation in the formal economy. At the same time, the response to income tax depends not only on its amount. Trust in state institutions and the quality of tax administration are also of significant importance. These institutional characteristics determine the extent to which changes in PIT affect the scale of the shadow economy in the countries of the region.

With this taken into account, below is a summary of statistics on income tax rates in Central European countries, allowing for a comparison of the tax burden on labour in 2024.

Tab. 6 Personal income tax in Central Europe (2024)

Country	Top marginal PIT rate (%)
Austria	55
Germany	47.5
Slovenia	39
Poland	27.6
Czechia	23
Slovak Republic	21.7
Hungary	15

Source: OECD Tax Database, 2024.

The data presented shows differences in income tax rates between Central European countries. Austria and Germany have high marginal PIT rates, while Slovenia and Poland have moderate income tax rates. In the Czech Republic and Slovakia, the PIT rate is relatively low, and Hungary stands out for its use of a flat rate, which reduces the progressivity of income tax.

Differences in income tax rates create unequal conditions for the tax burden on labour, however, they do not in themselves allow for an explicit assessment of the scale of the shadow economy. The impact of PIT may depend on institutional factors, including the level of trust in public institutions and the quality of tax administration, which may strengthen or weaken the response of economic agents to the tax burden. For a more comprehensive assessment of the tax burden on labour, social contributions must also be taken into account. As they also form a tax wedge and influence the decisions of employees and employers.

3.7 Social security contribution

Social security contributions are mandatory payments related to employment income and are paid by employees, employers, or both. They are usually calculated as a percentage of gross wages and therefore increase the overall cost of formal labour. Higher contribution rates enlarge the tax wedge and reduce net income from official employment. This can motivate both workers and firms to limit participation in the formal labour market, for example through undeclared work or partial wage payments outside the official system.

The structure of social security contributions is also important. In most Central European countries, a substantial part of contributions is paid by employers. Although these contributions are not directly visible to employees, they increase labour costs and are reflected in lower gross wages or reduced demand for labour. As a result, employer-paid contributions indirectly affect workers' incentives to participate in formal employment.

For example, a worker close to retirement age may face a situation where a large share of gross earnings is absorbed by social security contributions, while the expected increase in future pension benefits appears limited. In such a case, the worker may choose partially undeclared income in order to increase current disposable income, even at the cost of lower future pension entitlements. This example illustrates how high social security contributions can weaken incentives to fully participate in the formal labour market, particularly at later stages of working life.

The importance of social contributions as a determinant of the shadow economy is confirmed by numerous empirical studies. Taxes and social contributions explain approximately 35–52% of the variation in the size of the shadow economy, which exceeds the influence of government regulation, institutional quality, corruption, and enforcement measures (Schneider, Williams, 2013).

Tab. 7 Total social security contributions as a percentage of GDP (2024)

Tax category	Social security contributions (SSC)	Social security contributions (SSC) by employees	Social security contributions (SSC) by employers	Social security contributions (SSC) by self-employed or non-employed
Austria	15.6	6.3	7.3	2
Czechia	15.7	3.2	9.5	3
Germany	14.9	6.5	7	1.4
Hungary	10.1	6.3	3.8	0
Slovenia	16.9	8.5	5.8	2.5
Slovak Republic	15.6	3.7	9.2	2.7
Poland	14.1	0	5.1	9

Source: OECD, Revenue Statistics – Comparative tax revenue data, 2024.

Tab. 8 Social security contributions by employees and employers as a percentage of gross wages (Central European countries, 2024)

Country	Employee contribution (% of gross wage)	Employer contribution (% of gross wage)	Components covered by social security contributions
Germany	~20.0%	~20.0%	Pension insurance, Unemployment insurance, Health insurance, Long-term care insurance (split roughly 50/50; accident insurance paid fully by employer)
Czech Republic	7.10%	24.80%	Pension insurance, Sickness insurance, Unemployment policy (health insurance paid separately: 4.5% employee / 9% employer)
Austria	~18.1%	~21.0%	Pension insurance, Health insurance, Unemployment insurance, Accident insurance
Hungary	18.50%	13.00%	Pension insurance, Health insurance, Labour market contributions (unemployment, sickness benefits)
Poland	~22.7%	~22.1%	Old-age pension, Disability insurance, Sickness insurance, Accident insurance, Labour Fund
Slovakia	9.40%	25.20%	Pension insurance, Sickness insurance, Unemployment insurance, Guarantee fund, Accident insurance
Slovenia	22.10%	16.10%	Pension and disability insurance, Health insurance, Unemployment insurance, Parental protection

Source: Author's elaboration based on official national social security systems

A comparison of social security contributions calculated as a share of wages and as a share of GDP shows a persistently high labour burden in Central European countries. In Germany, Austria, Czechia, Slovakia, and Slovenia, social contributions account for around 15–17% of GDP, indicating a significant macroeconomic role of social security systems. At the same time, the contribution structure shows that employers bear a large part of the burden, which directly increases the cost of formal employment.

This combination of a high overall burden and its strong concentration on labour income reinforces the tax wedge and reduces the attractiveness of official employment. Workers and firms in these countries therefore become more sensitive to current labour costs, which increases incentives to use informal forms of employment and partially undeclared payments. In Poland and Hungary, where social security contributions represent a lower share of GDP, these incentives may be relatively weaker, although other institutional factors also play a role.

Overall, a high social security contribution burden affects not only short-term decisions regarding formal employment but also long-term participation in the official labour market. When contributions significantly reduce current income and future pension benefits are perceived as distant, incentives to remain fully in the formal sector weaken. In this context, retirement age becomes an important complementary factor, as it determines the expected length of formal employment and the balance between current contribution costs and future pension rights. The following section therefore examines how retirement age may influence the level of the shadow economy in the analysed countries.

3.8 Government effectiveness

Government effectiveness is an important institutional factor influencing incentives to participate in the formal or informal economy. It reflects the quality of public administration, policy implementation, regulatory enforcement, and the credibility of state institutions. In the context of the shadow economy, higher government effectiveness is generally associated with stronger compliance, more efficient tax administration, and higher trust in public institutions.

Government effectiveness is most commonly measured using the World Bank's Worldwide Governance Indicators (WGI), which aggregate data from multiple sources to capture perceptions of public service quality, administrative capacity, and the independence of the civil service from political pressures (Kaufmann, Kraay, Mastruzzi, 2010). The standardized nature of the indicator allows for meaningful cross-country comparisons.

Empirical studies typically find a negative relationship between government effectiveness and the size of the shadow economy, as stronger institutions reduce incentives to operate informally (Schneider, Enste, 2013; Buehn, Schneider, 2012). However, the role of government effectiveness is not uniform across countries and depends on national institutional arrangements and enforcement practices. In the-

ory, excessively strict regulation and administrative control may also increase compliance costs and reduce flexibility, potentially encouraging informal activity in a manner analogous to non-linear effects observed in tax compliance. In this context, government effectiveness constitutes part of the broader institutional environment within which labour market and pension-related policies, including regulations affecting retirement behaviour, operate.

3.9 Retirement age

As retirement approaches, formal employment incentives weaken, as additional years of formal work provide only a limited increase in future pension payments while maintaining a high tax burden and social contributions. In such conditions, formal employment becomes less attractive compared to informal employment. A similar mechanism may arise in the case of early retirement, when pension income is supplemented by informal employment. The basis for this assumption is that the high burden on labour reduces the attractiveness of formal employment, while the weak link between social contributions paid and future pension payments increases the incentives for workers to leave the formal sector (Schneider, 2011; OECD, 2019). At the same time, the direct relationship between retirement age, early retirement and the scale of the shadow economy has been studied to a limited extent in the current literature, which justifies the inclusion of this factor in further analysis.

Recent trends in Central European countries are characterised by a gradual increase in the retirement age in response to population ageing and the need to ensure the long-term sustainability of pension systems. According to forecasts, the average retirement age in EU countries will approach 67 by 2060, with increases planned in more than 20 countries (OECD, 2025). Central European countries generally follow this trend, but at different speeds and through different institutional mechanisms. In this regard, the table below provides a brief overview of the official retirement age in Central European countries. It should be noted that for several countries, the figures given are indicative, as the retirement age is determined by the year of birth and is gradually changing. The table reflects the standard retirement age without taking into account special benefits, such as having children, special professional regimes or other individual conditions. The pension systems of individual countries are discussed in more detail below.

Tab. 9 Statutory retirement age for men in selected Central European countries, 2010–2024 years

Year/Country	2010	2011	2012	2013	2014	2015	2016	2017	2018	2019	2020	2021	2022	2023	2024
Austria	65	65	65	65	65	65	65	65	65	65	65	65	65	65	65
Czechia	62,3	62,5	62,8	63,0	63,3	63,5	63,8	64,0	64,2	64,3	64,5	64,7	64,8	65,0	65,0
Germany	65,0	65,1	65,2	65,3	65,4	65,5	65,6	65,7	65,8	65,9	66,0	66,0	66,0	66,0	66,0
Hungary	62	62,5	63	63,5	64	64,5	64,5	64,5	65	65	65	65	65	65	65
Slovenia	63	63,5	64	65	65	65	65	65	65	65	65	65	65	65	65
Slovak Republic	62,0	62,0	62,0	62,0	62,0	62,0	62,0	62,2	62,4	62,6	62,8	63,0	63,2	63,4	63,6
Poland	65	65	65	65,4	65,5	65,75	66	66,25	65	65	65	65	65	65	65

Source: Author's compilation based on official national legislation and OECD data.

Tab. 10 Statutory retirement age for women in selected Central European countries, 2010–2024 years

Year/Country	2010	2011	2012	2013	2014	2015	2016	2017	2018	2019	2020	2021	2022	2023	2024
Austria	60	60	60	60	60	60	60	60	60	60	60	60	60	60	60,5
Czechia	61,3	61,5	61,8	62,0	62,3	62,5	62,8	63,0	63,2	63,3	63,5	63,7	63,8	64,0	64,0
Germany	65,0	65,1	65,2	65,3	65,4	65,5	65,6	65,7	65,8	65,9	66,0	66,0	66,0	66,0	66,0
Hungary	61	61,5	62	62,5	63	63,5	64	64,5	65	65	65	65	65	65	65
Slovenia	63	63,5	64	65	65	65	65	65	65	65	65	65	65	65	65
Slovak Republic	62,0	62,0	62,0	62,0	62,0	62,0	62,0	62,2	62,4	62,6	62,8	63,0	63,2	63,4	63,6
Poland	60	60	60	60,25	60,5	60,75	61	61,25	60	60	60	60	60	60	60

Source: Author's compilation based on official national legislation and OECD data.

Austria. The pension system of Austria is based on mandatory public pension insurance of a pay-as-you-go type. The statutory retirement age differs by gender: for men it is set at 65 years, while for women a lower age of 60 years applied until 2023. Since 2024, a gradual increase in the retirement age for women has begun, which will be completed by reaching the level of 65 years by 2033. At present, gender differentiation remains in the system, but it is being gradually eliminated within the framework of long-term pension reform. (OECD, 2024; BMSGPK, 2025)

Czechia. The pension system of Czechia is characterized by a high degree of institutional complexity. The retirement age is determined not by calendar year but by the year of birth of the insured person, which leads to significant differentiation. Historically, the system has also included gender differences, while women are granted additional benefits related to the number of children raised, allowing for earlier retirement. Recent reforms show a tendency toward convergence of the retirement age for men and women, as well as its gradual increase. (CSSZ, 2025)

Germany. The pension system of Germany is among the most stable in Europe. The retirement age is the same for men and women and is determined exclusively by the year of birth. As a result of implemented reforms, the retirement age is being gradually increased from 65 to 67 years, with the age of 67 applying to all individuals born in 1964 and later. The system does not include benefits related to gender or the number of children, which makes it relatively simple and transparent for economic analysis. (BMAS, 2025)

Hungary. During the 2010s, the pension system of Hungary underwent a period of gradual increases in the retirement age. At the beginning of the analyzed period,

the retirement age differed by gender; however, as a result of reforms, it was unified. Since the early 2020s, the retirement age has been set at 65 years for both men and women. Until this level was reached, the increase was implemented gradually, reflecting the transitional nature of the reforms. In its current form, the system is characterized by the absence of gender differences and a fixed statutory retirement age. (EURAXESS Hungary, 2025; OECD, 2023)

Slovenia. The pension system of Slovenia is based on the principle of gender equality in determining the retirement age. Since 2013, the statutory retirement age for both men and women have been set at 65 years, provided that the minimum insurance period is met. Despite the existence of early retirement mechanisms for individuals with long insurance careers, the official retirement age remains unchanged, which ensures relative stability of the system and reduces institutional uncertainty for economic analysis. (OECD, 2025; Republic of Slovenia, 2025)

Slovakia. The pension system of Slovakia combines elements of a fixed retirement age and an automatic adjustment mechanism. Until 2017, the retirement age was set at 62 years. Subsequently, an automatic mechanism for increasing the retirement age linked to changes in life expectancy was introduced, while an upper limit of 64 years was established. For the purposes of this analysis, the retirement age in Slovakia is treated as identical for men and women under standard conditions, without taking individual benefits into account. (Liga Pre Seniorov, 2025; OECD, 2023)

Poland. The pension system of Poland is characterized by persistent gender differentiation in the statutory retirement age. The retirement age is set at 65 years for men and 60 years for women. In the period 2013–2017, a reform was implemented that provided for a gradual increase in the retirement age to 67 years for both genders. However, in 2017 this reform was reversed, and the retirement age was returned to its previous levels, which makes the Polish case illustrative in terms of institutional instability of pension policy. (OECD, 2025; Republic of Poland, 2025)

Thus, in Central European countries the retirement age differs in terms of level, pace of change, and determination rules, as shown by the conducted overview of pension systems. These differences may affect the behavior of older workers and their participation in formal or informal employment. Although a direct relationship between retirement age and the size of the shadow economy is rarely examined in contemporary literature, the described institutional features allow for the assumption that such an effect may exist. Moreover, the retirement age is closely linked to migration, as changes in pension rules influence decisions on labor and post-retirement migration; therefore, the role of migration in shaping the shadow economy is examined in the following section.

3.10 Migration

In academic literature, migration is seen as a process that's closely connected to a person's age and how long they've been working. In general, migration is seen as a decision to improve living conditions, mainly through higher income and better

job opportunities. That's why age is one of the key factors in migration behaviour. As people approach retirement age, the economic benefits of moving decrease, while the costs and difficulties associated with migration increase. In the context of an ageing population, this leads to a decline in labour mobility and a reduction in migration flows.

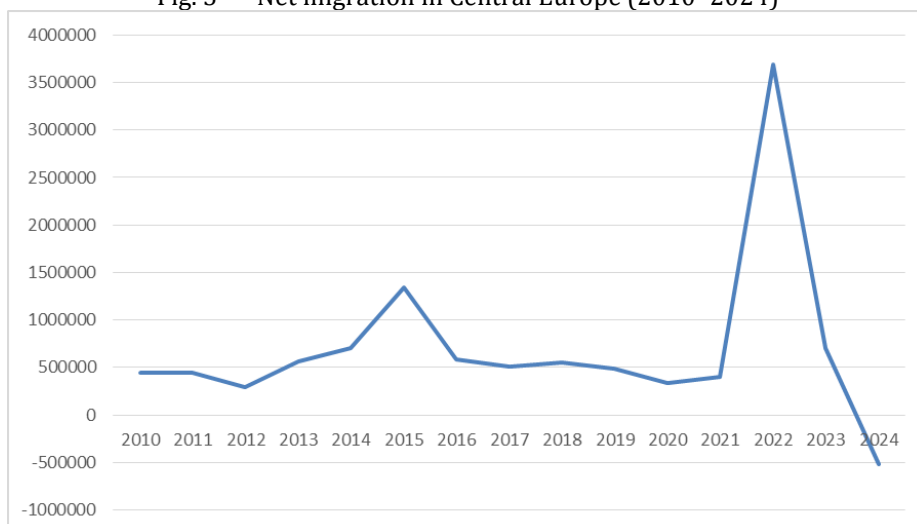
Young people migrate more often because they have a longer working life ahead of them and moving can bring tangible benefits in the future. In contrast, older workers are less likely to change their country of residence because the expected economic return from migration is lower for them and their ties to their current place of residence and social environment are stronger. Raising the retirement age increases the share of older workers in the labour market, which further limits the scale of labour migration and reduces the inflow of labour migrants, especially in countries with pronounced demographic ageing processes. (Zaiceva, 2014)

At the same time, the reduction in labour migration does not mean a complete decline in migration activity among older population groups. Literature distinguishes between return migration and retirement migration. Return migration involves migrants returning to their country of origin after completing their working careers. Retirement migration involves pensioners moving to other countries, usually with a lower cost of living, favourable climatic conditions or more accessible medical services. These forms of migration are less related to the labour market and are largely determined by quality-of-life factors. (Warnes 2009)

This study assumes that the link between migration and retirement age has broader economic implications. As shown in the previous section, many countries are experiencing a steady trend towards raising the retirement age, which, combined with an ageing population, may increase pressure on the labour market. This primarily affects low-wage sectors, where formal employment opportunities are becoming more limited. In such conditions, some migrants and older workers may move into informal employment if formal work becomes less accessible or less economically attractive. Similar effects are possible in the case of return or retirement migration, when, after moving or returning to the country of origin, mainly informal sources of income remain, especially when there are differences in pension and tax systems between countries. Thus, it is assumed that migration, combined with an increase in the retirement age, may indirectly affect the level of the shadow economy through changes in the labour market and institutional conditions.

For further analysis, this paper uses the net migration indicator, which reflects the difference between immigration and emigration in a country over a given period.

Fig. 5 Net migration in Central Europe (2010–2024)



Source: World Bank Data, author's own processing.

Data on net migration in Central European countries for the period 2010–2024 show that until 2021, total net migration in the region remained relatively stable at around 50,000 people per year. In 2010–2019, there were moderate fluctuations in the migration balance between countries, but without any sharp changes or a sustained upward or downward trend. The COVID-19 pandemic period was accompanied by a slight decline in migration activity, but overall, migration flows in the region remained limited and stable until 2021.

A sharp increase in net migration is observed in 2022, which is primarily due to the war in Ukraine and the mass migration of Ukrainian citizens to Central European countries. A significant part of these migration flows was forced and temporary in nature and was not related to the labour market in the classical sense. In subsequent years, there has been a decline in net migration, indicating a gradual stabilisation of the migration situation. This study assumes that such sharp migration surges may temporarily increase the burden on the labour markets and institutional systems of host countries and, under certain conditions, create an environment for the growth of informal employment. This justifies the use of net migration as a control variable in the further analysis of factors related to the dynamics of the shadow economy.

4 Methodology

This study analyses the possible impact of early retirement on the size of the shadow economy in Central European countries, including Austria, the Czech Republic, Germany, Hungary, Poland, the Slovak Republic, and Slovenia. The aim of the study is to identify the relationship between early retirement indicators and the level of the shadow economy, as well as to assess the potential tax losses associated with the transition of part of the population to informal employment and to measure the overall level of the shadow economy based on available statistical data for the period 2010–2024.

There is no universal method for measuring the shadow economy in scientific literature, due to the lack of a single definition of this phenomenon and its unobservable nature. Therefore, different methodological approaches are used in empirical studies. The most common are direct methods based on surveys and tax audits (Feld and Larsen, 2005) and indirect macroeconomic methods using indicators of demand for cash or discrepancies in national accounts (Tanzi, 1983; Schneider and Enste, 2000). These approaches allow the scale of the shadow economy to be estimated but are based on a number of assumptions. Alongside them, model approaches are used, in which the shadow economy is estimated indirectly on the basis of a set of observable factors and indicators. Within this group of methods, the MIMIC (Multiple Indicators Multiple Causes) method is widely used in empirical studies.

The MIMIC method was first applied to the analysis of the shadow economy in the works of Frey and Weck-Hannemann (1984) and was then widely developed in the studies of Giles (1999), Schneider and Enste (2000), Bühn and Schneider (2008), and many other authors. In this approach, the shadow economy is treated as a latent variable, i.e., a value that cannot be observed directly but can be estimated through a system of causal factors and observable indicators. The research literature emphasises that the specification of the MIMIC model is not fixed and depends on the research objective, time period, and data availability.

This study is a quantitative empirical analysis based on secondary macroeconomic data. The main tool used is the MIMIC model, adapted to take into account the research objectives and available data.

The model is evaluated using Stata 19 software with the SEM (structural equation modelling) module.

The shadow economy is represented in the model as a latent variable SE (shadow economy). The model consists of two parts: a structural equation (causes → SE) and measurement equations (SE → indicators).

Structural part:

For each country, the assessment is based on annual observations. Let t be the year, then the structural part is given by the equation:

$$SE_t = \alpha + \beta_1 GDPpc_t + \beta_2 SELFEMP_t + \beta_3 UNEMP_t + \beta_4 GOVEFF_t + \beta_5 PIT_t + \beta_6 CIT_t + \beta_7 SSC_t + \beta_8 + \beta_9 RA_w_t + \beta_{10} NETMIG_t + \varepsilon_{1t}$$

Where:

- SE_t — latent (unobservable) level of the shadow economy,
- GDP_{pc_t} — GDP per capita,
- $SELFEMP_t$ — self-employment,
- $UNEMP_t$ — unemployment,
- $GOVEFF_t$ — government effectiveness,
- PIT_t — personal income tax,
- CIT_t — corporate income tax,
- SSC_t — social security contributions,
- RA_{mt}/RA_{wt} — early retirement indicators (men/women),
- $NETMIG_t$ — net migration,
- ε_{1t} — structural equation error.

The sign of the coefficient β_n shows the direction of the factor's relationship with the latent level of the shadow economy, and statistical significance is assessed by p-value.

In addition, to ensure correct model specification and comparability of estimates, preliminary adjustment of variables and model parameters is used. In particular, the gross domestic product per capita indicator is used in logarithmic form in a number of specifications (LogGDP_{pc_t})

Logarithmisation reduces the influence of variable scale, smooths out possible outliers, and contributes to the stability of estimates in models with latent variables.

Measurement part

The latent variable SE_t manifests itself through observable indicators. Two indicators are used in this diagram: GDP growth ($GDPg$) and labour force participation (LFP).

$$GDPg_t = \gamma_1 SE_t + \varepsilon_{2t}$$

$$LFP_t = \gamma_2 SE_t + \varepsilon_{3t}$$

Where:

- $GDPg_t$ — GDP growth,
- LFP_t — labor force participation,
- γ_1, γ_2 — factor loadings,
- $\varepsilon_1, \varepsilon_2$ — measurement errors.

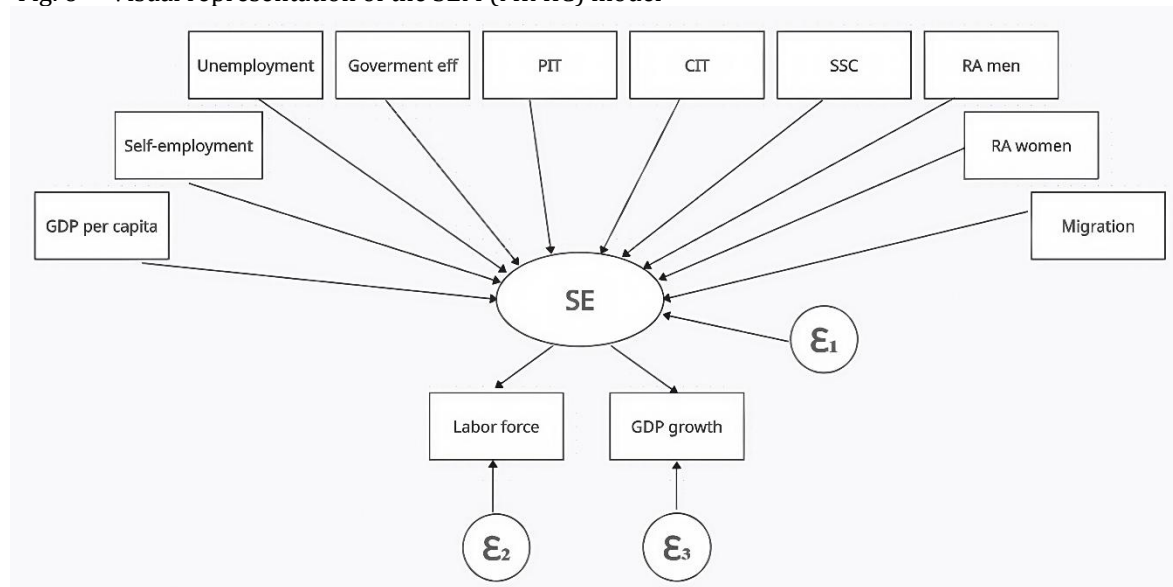
Since SE_t is not observed and does not have its own measurement scale, one of the coefficients in the measurement part is fixed to identify the model. In our model, the load factor $\gamma_1 = 1$ (constrained).

The parameters of the MIMIC model are estimated using the Maximum Likelihood method implemented in the SEM module. The estimation process uses information about the covariance structure of the observed variables, i.e., how these variables are related to each other, which ensures a consistent estimation of both the structural and measurement parts of the model.

The interpretation of the model results is based on the analysis of the coefficients of the structural part. Coefficients with a p-value below 0.05 are considered statistically significant and are regarded as factors influencing the latent level of the shadow economy. The sign of the coefficient indicates the direction of the relationship: a positive value means that the shadow economy grows as the corresponding factor increases, while a negative value indicates the opposite relationship. As the shadow economy is a latent variable, the coefficients are used to analyse the direction and relative significance of the influence and are not interpreted in absolute units or percentages.

Evaluation of the MIMIC model is carried out in stages. The first stage uses an extended specification that includes a full set of potential factors, including early retirement indicators, which are key to the research objective.

Fig. 6 Visual representation of the SEM (MIMIC) model



In the second stage, the model can be refined by excluding statistically insignificant variables if their presence does not add any additional information and complicates the interpretation of the results. Retaining such variables may reduce the stability of estimates and affect the significance of other coefficients. The refined specification allows for a clearer identification of the main drivers of the shadow economy while maintaining theoretically sound control factors.

The analysis is performed separately for each Central European country. This allows taking into account national characteristics of the economic structure, institutional environment and labour market, as well as identifying cross-country differences in factors.

5 Data

The analysis in this study is based on data covering the period from 2010 to 2024. This time interval is considered the most relevant for examining the selected Central European countries. The sample includes Austria, the Czech Republic, Germany, Hungary, Poland, the Slovak Republic, and Slovenia. The chosen period makes it possible to focus on contemporary economic and institutional conditions and to avoid distortions related to earlier global economic shocks, such as the oil crises of the 1970s, the recession of the early 1990s, the dot-com crisis, and the global financial crisis of 2008–2009. The period analyzed is also characterized by a high degree of data availability and cross-country comparability.

The main part of the data was collected from the World Bank Data database and includes indicators of gross domestic product (GDP growth, GDP per capita), the level of self-employment for the period 2010–2023, unemployment, migration, government effectiveness estimates, and labor force participation. Additional and missing indicators, in particular data on corporate income tax, personal income tax, social security contributions, as well as self-employment for the year 2024, were obtained from the OECD database. Information on retirement age was collected from official OECD publications and national governmental sources. In all cases, the standard statutory retirement age was used, without taking into account special allowances such as child-related benefits, occupational regimes, or other individual conditions. This approach ensures a more generalized and standardized perspective, which is particularly important given the institutional complexity of pension systems and differences in retirement age regulations across the countries analyzed.

Below is a brief description of all variables used in the analysis and their main characteristics.

- **GDP growth**

This indicator reflects the annual rate of change in gross domestic product and is used to account for business cycle phases and short-term macroeconomic fluctuations. The data are reported on an annual basis and expressed in percentage terms, which allows for comparisons of economic dynamics across countries and overtime.

- **GDP per capita**

This indicator is used as a proxy for the level of economic development and income of the population. The data are reported annually and expressed in current U.S. dollars.

- **Self-employment**

This variable reflects the share of self-employed individuals in total employment. The data are reported annually and expressed as a percentage of total employment.

- **Government effectiveness**

This indicator captures the quality of public governance, including the effectiveness of public institutions, the quality of public services, and the government's capacity to formulate and implement policies. The data are provided in the form of

a standardized index with a fixed scale and are used to account for institutional differences across countries.

- **Unemployment**

This indicator reflects the share of unemployed individuals in the labor force. The data are reported annually and expressed as a percentage of the total labor force.

- **Corporate income tax**

This variable reflects the statutory corporate income tax rate and is used as a measure of the tax burden on businesses. The data are reported annually and expressed in percentage terms.

- **Personal income tax**

This indicator reflects the statutory top marginal personal income tax rate and is used to assess potential incentives for income shifting into the shadow economy. The data are expressed in percentage terms.

- **Social security contributions**

This variable captures the level of mandatory social security contributions related to employment and is used as a component of the overall tax burden on labor. The data are expressed in percentage terms.

- **Retirement age**

This indicator reflects the official statutory retirement age, reported separately for men and women, without accounting for special allowances or individual conditions. The data are reported on an annual basis.

- **Net migration**

This indicator reflects the migration balance, defined as the difference between the number of immigrants and emigrants during a given year, and is used to account for demographic and labor market shifts within a country. The data are reported annually and expressed in absolute numbers.

- **Labor force participation**

This indicator reflects the share of the working-age population participating in the labor market, including both employed and unemployed individuals. The data are reported annually and expressed as a percentage of the relevant population group.

Overall, the data used provides a comparable and robust empirical basis for the subsequent analysis.

6 Results

6.1 Austria case

Tab. 11 Effect of Retirement Age on the Size of the Shadow Economy (Austria)

	OIM		z	P> z	[95% conf. interval]	
	Coefficient	std. err.				
Structural						
L1						
RA_m	.8983889	64.70424	0.01	0.989	-125.9196	127.7164
RA_w	-6.991063	65.6398	-0.11	0.915	-135.6427	121.6606

The model results show that the retirement age does not have a statistically significant impact on the size of the shadow economy in Austria. This result is related to the institutional features of the country's pension system. Between 2010 and 2024, the retirement age remained practically unchanged: 65 for men and 60 for women. It was only in 2024 that a law came into force to gradually raise the retirement age for women to 65, with the increase being implemented in stages — six months per year. As a result, the actual changes in the retirement age were minimal during the period under review, which explains the lack of a noticeable impact of this factor on the dynamics of the shadow economy.

Tab. 12 Structural and Measurement Estimates of the SE (Austria)

	OIM		z	P> z	[95% conf. interval]	
	Coefficient	std. err.				
Structural						
L1						
LogGDPpc	16.85785	9.285015	1.82	0.069	-1.340447	35.05614
SELFEMP	-5.87147	2.949739	-1.99	0.047	-11.65285	-.0900874
UNEMP	-2.630426	2.724937	-0.97	0.334	-7.971204	2.710352
CIT	5.782372	5.179436	1.12	0.264	-4.369137	15.93388
PIT	.0106856	.2720077	0.04	0.969	-.5224398	.5438109
SSC	-11.69388	5.344781	-2.19	0.029	-22.16946	-1.218299
Measurement						
GDP_g						
L1						
_cons	1 (constrained)					
	-40.08009	.	.	.	.	.
LFP						
L1						
_cons	.014687	.0024336	6.04	0.000	.0099172	.0194567
	60.38587	.	.	.	.	.
var(e.GDP_g)	20.83452	.			.	.
var(e.LFP)	.1467488	.0555853			.0698486	.3083126
var(e.L1)	-1.00e-09	.			.	.

In the second stage of modelling (final specification), statistically insignificant variables were excluded. As a result, the key drivers of the shadow economy are self-employment and social security contributions. Both factors have a statistically significant negative impact on the size of the shadow economy, which means that an increase in these factors in Austria is accompanied by a reduction in the scale of the shadow economy. In Austria, self-employment is a legally recognised status that requires mandatory registration, payment of taxes and social contributions, as well as administrative control and heavy fines in case of violations. In addition, self-employment in Austria is predominantly represented by highly skilled and high-income professions. Similarly, social security contributions are directly linked to pension rights, health insurance and social protection, and evading them leads to the loss of social guarantees and fines, which makes formal employment more profitable and reduces incentives to work in the shadow economy.

Other factors, including the unemployment rate, income and corporate tax rates, and the logarithm of GDP per capita, did not show a statistically significant impact on the size of the shadow economy in Austria during the period under review. The lack of significance of these variables indicates that, in conditions of stable income levels and developed institutions, their impact on the dynamics of the shadow economy is limited.

6.2 Czechia case

Tab. 13 Effect of Retirement Age on the Size of the Shadow Economy (Czechia)

	OIM		z	P> z	[95% conf. interval]	
	Coefficient	std. err.				
Structural						
L1						
RA_m	-8.146376	602.7176	-0.01	0.989	-1189.445	1173.164
RA_w	1.259808	611.8738	0.00	0.998	-1197.991	1200.51

The results of the model indicate that the retirement age does not have a statistically significant effect on the size of the shadow economy in Czechia. This outcome should be interpreted in light of the institutional specificity of the Czech pension system, which is characterized by a high degree of complexity and differentiation. In Czechia, the retirement age is determined not by the calendar year but by the year of birth of the insured individual and has historically included gender differences as well as additional benefits for women depending on the number of children raised. Within the framework of this study, pension-related variables were aggregated and represented in a simplified form, which may have smoothed the effects of individual institutional mechanisms. A more detailed analysis of differentiated pension rights and benefits may therefore represent a separate direction for future research.

Tab. 14 Structural and Measurement Estimates of the SE (Czechia)

	OIM					
	Coefficient	std. err.	z	P> z	[95% conf. interval]	
Structural						
L1						
GDP_pc	-.000042	.000321	-0.13	0.896	-.0006711	.0005871
SELFEMP	-1.967458	1.132729	-1.74	0.082	-4.187566	.2526498
UNEMP	1.285295	.7905285	1.63	0.104	-.264112	2.834703
CIT	1.03531	.9688954	1.07	0.285	-.8636902	2.93431
PIT	2.517783	.6762124	3.72	0.000	1.192431	3.843135
SSC	-3.700902	1.38469	-2.67	0.008	-6.414844	-.9869588
Measurement						
GDP_g						
L1	1	(constrained)				
_cons	10.23938	.	.	.	.	.
LFP						
L1	.1059188	.0031859	33.25	0.000	.0996746	.1121631
_cons	60.81737	.	.	.	.	.
var(e.GDP_g)	2.979861	.			.	.
var(e.LFP)	.4538592	.			.	.
var(e.L1)	-1.00e-09	.7711335			.	.

In the second stage of modelling, personal income tax and social security contributions were identified as key drivers of the shadow economy. Both factors have a statistically significant impact on the size of the shadow economy, but their effects differ in direction. Personal income tax has a positive effect. This indicates that the tax burden on labour may create incentives for part of the income to be transferred to the informal sector, especially among workers with lower and middle incomes, for whom the tax wedge is more noticeable.

Social security contributions, by contrast, exert a statistically significant negative effect on the size of the shadow economy, similarly to the results obtained for Austria. In the Czech context, this effect reflects the institutional linkage between social rights and participation in the formal social insurance system.

Other factors, including the unemployment rate, income per capita, the corporate income tax rate, and self-employment, do not demonstrate a statistically significant effect on the size of the shadow economy over the analyzed period. This suggests that under conditions of a relatively stable institutional environment and labor market, their influence on the shadow economy is limited and secondary to tax and social incentives.

6.3 Germany case

Tab. 15 Effect of Retirement Age on the Size of the Shadow Economy (Germany)

	OIM					
	Coefficient	std. err.	z	P> z	[95% conf. interval]	
Structural						
L1						
RA_m	42.9357	.	.	.	.	.
RA_w	-.762486	29.32536	-0.03	0.979	-58.23914	56.71416

The results show that the retirement age does not have a statistically significant impact on the size of the shadow economy in Germany. As there is no gender differentiation, the variables reflecting the retirement age of men and women are linearly dependent, and the appearance of dots in the assessment results indicates that this parameter is actually assessed as a single institutional characteristic rather than two separate factors. Despite the reforms carried out between 2010 and 2019, which provided for a gradual increase in the retirement age from 65 to 67, these changes were gradual and predictable during the period under review. As a result, the retirement age is not an independent driver of the shadow economy in Germany.

Tab. 16 Structural and Measurement Estimates of the SE (Germany)

	OIM					
	Coefficient	std. err.	z	P> z	[95% conf. interval]	
Structural						
L1						
LogGDPpc	6.087463	2.78086	2.19	0.029	.6370783	11.53785
SELFEMP	-2.398985	1.040061	-2.31	0.021	-4.437466	-.360503
GOVEFF	14.64825	4.24105	3.45	0.001	6.335945	22.96056
UNEMP	2.298995	.6419167	3.58	0.000	1.040862	3.557129
SSC	-7.860026	1.829308	-4.30	0.000	-11.4454	-4.274649
NETMIG	7.96e-06	2.99e-06	2.66	0.008	2.09e-06	.0000138
Measurement						
GDP_g						
L1	1	(constrained)				
_cons	7.484299	.	.	.	.	.
LFP						
L1	-.2434273	.0062495	-38.95	0.000	-.255676	-.2311785
_cons	58.47697	.	.	.	.	.
var(e.GDP_g)	1.201702	.			.	.
var(e.LFP)	.3546403	.			.	.
var(e.L1)	-1.00e-09	.2906642			.	.

In the final stage of modelling, the logarithm of GDP per capita, the efficiency of public administration, the unemployment rate, net migration, self-employment and social contributions have a statistically significant impact on the size of the shadow

economy. At the same time, the logarithm of GDP per capita, the efficiency of public administration, unemployment, and net migration demonstrate a direct effect, while self-employment and social contributions have the opposite effect.

The positive effect of the logarithm of GDP per capita indicates the sensitivity of the shadow economy to changes in income levels, even in a highly developed economy. The effectiveness of public administration demonstrates one of the strongest effects, emphasising the importance of institutional quality and trust in public institutions. The statistically significant impact of unemployment reflects the link between labour market tensions and informal economic activity. Net migration, despite its small effect, also has an impact, which may be related to the record migration flows of recent years and their short-term impact on the labour market.

Self-employment and social contributions have a statistically significant negative impact on the size of the shadow economy, which is associated with the high degree of formalisation of self-employment and the institutional link between social rights and participation in the mandatory insurance system, similar to the results for Austria.

Consequently, in Germany, the shadow economy depends primarily on institutional and labour factors rather than on the pension system parameters.

6.4 Hungary case

Tab. 17 Effect of Retirement Age on the Size of the Shadow Economy (Hungary)

	OIM		z	P> z	[95% conf. interval]	
	Coefficient	std. err.				
Structural						
L1						
RA_m	3.369535	4.271969	0.79	0.430	-5.00337	11.74244
RA_w	-9.494764	4.648133	-2.04	0.041	-18.60494	-.3845897

The results show different effects of retirement age on the size of the shadow economy by gender. The retirement age of men is not statistically significant, while the retirement age of women has a statistically significant negative effect. This result can be explained by the specific features of the pension system, in particular the special programme Nők 40 (Women 40) programme. This programme allows women to retire after 40 years of official employment regardless of their calendar age (Act LXXXI, 1997). Under this system, only formal employment counts towards pensionable service, while informal employment does not create pension rights. This creates strong incentives for women to maintain their official employment status and limits the attractiveness of shadow work, as the loss of official service directly reduces the possibility of receiving a pension. As a result, the increase in the retirement age for women in Hungary is accompanied by a reduction in the scale of the shadow economy, which reflects the identified negative effect.

Tab. 18 Structural and Measurement Estimates of the SE (Hungary)

	OIM					
	Coefficient	std. err.	z	P> z	[95% conf. interval]	
Structural						
L1						
GDP_pc	.002917	.0004415	6.61	0.000	.0020517	.0037823
GOVEFF	89.12945	10.70931	8.32	0.000	68.1396	110.1193
UNEMP	-2.735216	.4813502	-5.68	0.000	-3.678645	-1.791787
CIT	1.490833	.2668365	5.59	0.000	.967843	2.013823
PIT	5.024974	1.241169	4.05	0.000	2.592327	7.457621
SSC	6.706555	1.125005	5.96	0.000	4.501586	8.911525
RA_w	-5.273322	.7412432	-7.11	0.000	-6.726132	-3.820512
NETMIG	.0007655	.0001285	5.96	0.000	.0005136	.0010173
Measurement						
GDP_g						
L1	1	(constrained)				
_cons	20.186	.	.	.	.	.
LFP						
L1	.2259433	.0758473	2.98	0.003	.0772853	.3746013
_cons	61.45009	.	.	.	.	.
var(e.GDP_g)	1.300005	.			.	.
var(e.LFP)	12.23402	4.849771			5.625212	26.60724
var(e.L1)	-1.00e-09	.2818163			.	.

In the final model specification, all included structural variables are statistically significant, because in all cases the p-value in the table is rounded to 0.000, which means $p < 0.005$.

GDP per capita and net migration demonstrate a positive and significant impact on the size of the shadow economy. However, their contribution is less pronounced compared to institutional and fiscal factors. Growth in per capita income may be accompanied by a complication of economic and tax relations, which expands opportunities for informal activity. The positive effect of migration indicates a higher probability of concentration of shadow employment among migrants, especially in sectors with low barriers to entry.

The variable of government effectiveness has one of the strongest positive effects, allowing it to be considered a key institutional factor in a given country. Improving the effectiveness of public administration, accompanied by tighter regulation and control, may encourage some economic activity to move into the informal sector.

Personal income tax and corporate income tax have a direct positive impact on the size of the shadow economy. This result is consistent with theoretical expectations and reflects increased incentives for tax evasion as the tax burden rises. A similar effect is observed for social security contributions, which increase the cost of formal employment and thus expand the informal sector.

Unlike other factors, the unemployment rate has a negative impact on the shadow economy in Hungary. This is because informal employment in the country

is mainly concentrated in labour-intensive sectors such as construction, services and small businesses. Rising unemployment means a reduction in labour demand and a decline in household income, which limits opportunities for informal employment. As a result, an increase in unemployment leads to a reduction in the shadow sector rather than its expansion.

6.5 Poland case

Tab. 19 Effect of Retirement Age on the Size of the Shadow Economy (Poland)

	OIM		z	P> z	[95% conf. interval]	
	Coefficient	std. err.				
Structural						
L1						
RA_m	-27.56583	44.76456	-0.62	0.538	-115.3028	60.17109
RA_w	27.72238	43.69738	0.63	0.526	-57.92292	113.3677

The results of the model for Poland show that the retirement age does not have a statistically significant impact on the size of the shadow economy for either men or women. This is despite changes in the retirement age associated with the 2013–2017 reform. The absence of a significant effect indicates that these changes in pension parameters did not create sustainable behavioural incentives that could influence the population's participation in the shadow economy.

Tab. 20 Structural and Measurement Estimates of the SE (Poland)

	OIM		z	P> z	[95% conf. interval]	
	Coefficient	std. err.				
Structural						
L1						
LogGDPpc	2.292077	4.164853	0.55	0.582	-5.870884	10.45504
SELFEMP	-2.894794	1.2483	-2.32	0.020	-5.341418	-.4481704
GOVEFF	21.01674	8.907069	2.36	0.018	3.559208	38.47428
CIT	.6416771	3.219387	0.20	0.842	-5.668206	6.95156
PIT	4.681636	5.000762	0.94	0.349	-5.119678	14.48295
SSC	-3.834507	1.682605	-2.28	0.023	-7.132352	-.5366629
Measurement						
GDP_g						
L1	1	(constrained)				
_cons	14.84585	.	.	.	.	.
LFP						
L1	.1028431	.0321506	3.20	0.001	.039829	.1658572
_cons	58.32899	.	.	.	.	.
var(e.GDP_g)	2.939908	.			.	.
var(e.LFP)	.9667135	.3795207			.4478429	2.086747
var(e.L1)	-1.00e-09	1.063495			.	.

In the second stage of modelling, three statistically significant factors determining the size of the shadow economy were identified. Self-employment, as in the cases of Germany and Austria, has a negative impact. This result indicates a relatively high level of formalisation of self-employment and limited opportunities for non-formal employment in this segment. The effectiveness of public administration demonstrates a statistically significant positive effect, which emphasises the important role of institutional factors in shaping the scale of the shadow economy in the selected country, as well as the dependence of informal activity on the level of trust in public institutions.

Social security contributions also have a negative impact on the size of the shadow economy. This result can be explained by the institutional tie between social rights and participation in the formal social security system. As a result, formal employment becomes relatively more attractive than working in the shadow economy.

Other factors, including the logarithm of GDP per capita, personal income tax rates and corporate income tax rates, did not have a significant impact. Accordingly, in the Polish context, institutional characteristics and the formalisation of employment play a key role in influencing the informal sector, while macroeconomic and fiscal parameters are of secondary importance.

6.6 Slovakia case

Tab. 21 Effect of Retirement Age on the Size of the Shadow Economy (Slovakia)

	OIM				
	Coefficient	std. err.	z	P> z	[95% conf. interval]
Structural					
L1					
RA_m	13.34518	9.976239	1.34	0.181	-6.207891 32.89825
RA_w	-2.011583	.	.	.	.

The retirement age does not have a statistically significant impact on the size of the shadow economy in Slovakia. Since the retirement age in the country is the same for men and women, this leads to a linear relationship between the relevant variables and explains the appearance of points in the assessment results. The lack of a significant effect is explained by the fact that in Slovakia, the retirement age was fixed at 62 until 2017, and subsequent changes were automatic and predictable, with an upper limit of 64, which did not create any sharp incentives to change labour behaviour and participate in the shadow economy.

Tab. 22 Structural and Measurement Estimates of the SE (Slovakia)

	OIM					
	Coefficient	std. err.	z	P> z	[95% conf. interval]	
Structural						
L1						
GDP_pc	.008561	.004277	-2.00	0.045	-.0169438	-.0001782
SELFEMP	1.543568	2.265208	0.68	0.496	-2.896157	5.983294
UNEMP	.2344327	.9942237	0.24	0.814	-1.71421	2.183075
CIT	-3.223643	1.905206	-1.69	0.091	-6.957779	.5104924
PIT	4.825519	2.040684	2.36	0.018	.8258516	8.825187
SSC	-1.904436	3.125082	-0.61	0.542	-8.029483	4.220611
Measurement						
GDP_g						
L1	1	(constrained)				
_cons	4.970436	.	.	.	.	.
LFP						
L1	-.3820418	.4782604	-0.80	0.424	-1.319415	.5553314
_cons	58.98344	1.327402	44.44	0.000	56.38178	61.5851
var(e.GDP_g)	4.570743	.			.	.
var(e.LFP)	.5972715	.5515345			.097758	3.649146
var(e.L1)	-1.00e-09	.			.	.

As for the final drivers affecting the size of the shadow economy, the final stage of modelling revealed a statistically significant positive relationship between personal income tax and the scale of the shadow economy. In the Slovak context, this result may be related to the gradual increase in the tax burden on labour between 2010 and 2024. In a relatively stable labour market and in the absence of sharp institutional changes, personal income tax represents the most visible element of tax pressure, directly affecting workers' disposable income and creating incentives to hide part of their income or use informal forms of remuneration.

In Slovakia, GDP per capita has a statistically significant but very small effect on the shadow economy. Compared to personal income tax, its influence is clearly weaker and economically negligible. This suggests that changes in overall income levels have only a limited impact on informal activity, while tax pressure on labour plays a much more important role in shaping the size of the shadow economy.

Overall, the results for Slovakia indicate the dominant role of direct tax incentives, with pension parameters and other macroeconomic factors having no significant impact on the size of the informal economy.

6.7 Slovenia case

Tab. 23 Effect of Retirement Age on the Size of the Shadow Economy (Slovenia)

	OIM				
	Coefficient	std. err.	z	P> z	[95% conf. interval]
Structural					
L1					
RA_m	-1.325664	1.427816	-0.93	0.353	-4.124131 1.472804
RA_w	.1529637	.	.	.	.

The retirement age does not have a statistically significant impact on the size of the shadow economy in Slovenia. In the model, the retirement age is considered to be the same for men and women, and therefore the programme identifies and evaluates this indicator as a single parameter. The absence of a significant effect is consistent with the structural characteristics of the Slovenian pension system, where pension parameters are relatively stable and do not cause sharp changes in labour market incentives during the period under review.

Tab. 24 Structural and Measurement Estimates of the SE (Slovenia)

	OIM				
	Coefficient	std. err.	z	P> z	[95% conf. interval]
Structural					
L1					
logGDPpc	1.599751	2.67318	0.60	0.550	-3.639586 6.839088
UNEMP	-.3849492	.3835703	-1.00	0.316	-1.136733 .3668348
CIT	1.145204	.9014509	1.27	0.204	-.6216074 2.912015
PIT	.8109579	.5904367	1.37	0.170	-.3462768 1.968193
SSC	-3.552688	1.533284	-2.32	0.021	-6.557869 -.5475069
Measurement					
GDP_g					
L1	1	(constrained)			
_cons	20.0592	.	.	.	.
LFP					
L1	.1023995	.0096542	10.61	0.000	.0834776 .1213213
_cons	59.92071	.	.	.	.
var(e.GDP_g)	5.532929	6.34597			.58435 52.38865
var(e.LFP)	.3908343	.157653			.1772715 .8616811
var(e.L1)	-1.00e-09	5.306129			.

It should be noted that Slovenia is the smallest country in the sample in both economic and geographical terms, which complicates the identification of stable factors affecting the shadow economy. Under the given conditions, the formation of a correct model specification required the reduction and combination of factors in the

structural part of the SEM model. Despite this, the final specification allowed us to identify one stable and statistically significant factor.

The results of the final model show that social security contributions have a negative impact on the size of the shadow economy in Slovenia. This can be explained by the high level of social insurance coverage and the strict ties between access to pension and health rights and participation in the formal employment system.

Other macroeconomic factors, including GDP per capita, unemployment, personal income and corporate income taxes, did not show a statistically significant impact on the size of the shadow economy.

7 Discussion

To discuss the results of this study, it is reasonable to return to the original objectives of the work and consider them in the appropriate order.

7.1 Early Retirement and the Shadow Economy

The initial assumptions regarding the impact of retirement age and early retirement on the shadow economy were based on the idea that, as individuals approach retirement age, incentives for formal employment weaken, while informal employment becomes relatively more attractive. This is related to the limited economic return from additional years of formal employment under a high tax and social contribution burden. It was also assumed that early retirement may encourage the combination of pension income with informal employment. Furthermore, differences in statutory retirement age, the pace of its increase, and institutional rules across Central European countries were expected to affect workers' behaviour in different ways. Finally, the possibility of an indirect effect of pension parameters on the shadow economy through migration decisions was considered, which justified the inclusion of migration in the subsequent analysis.

The results of the study indicate that, in all selected countries except Hungary, retirement age did not have a statistically significant impact on the size of the shadow economy over the period 2010–2024. This finding does not fully correspond with the conclusions of some previous empirical studies, including Chabak (2023). At the same time, it is worth noting that the relationship between retirement age and the shadow economy remains relatively underexplored, particularly in the context of Central European countries, underscoring the need for further empirical research in this area.

The absence of a statistically significant effect in most countries may be explained by the gradual and moderate nature of retirement age reforms in Central Europe, which took place within relatively stable pension systems. Although most OECD countries have followed a general trend towards increasing retirement age, these changes were not accompanied by noticeable shifts in the scale of informal economic activity. This suggests that pension parameters alone have limited explanatory power as independent determinants of the shadow economy.

Particular attention should be paid to the case of Hungary, where the retirement age of women demonstrated a statistically significant negative relationship with the size of the shadow economy. At the beginning of the analysed period, the statutory retirement ages of men and women differed; however, as a result of successive pension reforms, they were gradually equalised by the early 2020s. For women, these changes were more pronounced, which may explain why female retirement age emerged as a significant factor in the Hungarian model.

At the same time, this result should be interpreted in light of specific institutional features of the Hungarian pension system. Hungary operates the Women 40 programme, which grants women the right to retire after 40 years of

eligibility period regardless of statutory retirement age, potentially strengthening incentives to remain in formal employment (Act LXXXI, 1997). By contrast, in Slovenia, early retirement after a long insurance period is also possible, but it is accompanied by permanent benefit reductions if the statutory retirement age is not reached (Pension and Disability Insurance Act, ZPIZ-2, 2012). This difference in institutional rules may partially explain why a similar effect is not observed in the Slovenian case and indirectly confirms the importance of national pension regimes.

It should also be taken into account that in several countries, special provisions and preferential conditions for women's retirement exist, including those related to the number of children raised. However, these features of pension systems were not included in the empirical model, as their consideration would have significantly complicated the analysis and required detailed individual-level data.

The hypothesis concerning the relationship between retirement age, migration, and the shadow economy was also not empirically confirmed. Although migration exhibited a positive association with the size of the shadow economy in Germany and Hungary, this effect most likely reflects economic and political drivers of migration rather than a direct or indirect dependence on pension parameters.

7.2 Critical Discussion of Key Determinants of the Shadow Economy

To provide a comparative overview of the determinants of the shadow economy in Central Europe over the analysed period, the frequency with which individual factors exhibit a statistically significant effect is summarised in the figure below.

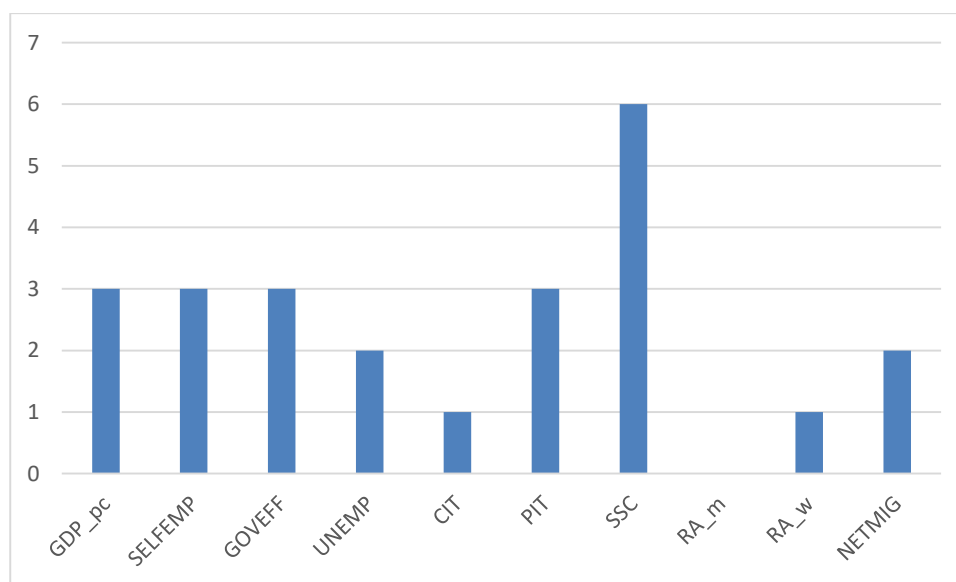


Fig. 7 Frequency of statistically significant determinants of the shadow economy in Central European countries (2010–2024)

As can be observed, social security contributions proved to be the most frequently identified and statistically significant factor influencing the level of the shadow economy, being significant in all countries except Slovakia. At the same time, in five out of six countries, an increase in social security contributions was accompanied by a reduction in the scale of the shadow economy, which does not correspond to the initial hypotheses and the conclusions of most of the literature that considers this indicator as an element of the tax wedge (Schneider, Williams, 2013). However, this result may reflect a high degree of population dependence on the social protection system, which includes pensions, health insurance, and social transfers, access to which is largely ensured through formal employment. Recent empirical studies focusing on European Union countries indicate that in states with a developed welfare state model, social security contributions may perform not only a fiscal but also an institutional function, thereby reducing incentives to move into the shadow economy (Mara, 2021). The results obtained for Austria, Germany, the Czech Republic, Poland, and Slovenia may reflect this mechanism, whereas in Hungary, the classic positive relationship between the level of social security contributions and the size of the shadow economy persists.

It then seems appropriate to consider the factor of self-employment. The initial hypotheses were that the impact of self-employment on the shadow economy is not uniform and depends on the level of economic development and the characteristics of the country's economic organisation. It was assumed that self-employment could perform both a "forced" and a "voluntary" function, with these forms being linked to the informal economy in different ways (Hart, 1973; Taylor, 1996; Maloney, 1998). The results obtained for Germany, Austria and Poland demonstrated a stable inverse effect of self-employment on the shadow economy. This result can be explained by the high level of economic development and relative stability of business regulations in these countries, where self-employment is largely voluntary and formal in nature and is considered a legal form of entrepreneurial activity (Schneider, 2012). In such conditions, the growth of self-employment does not reflect the expansion of the informal sector but, on the contrary, may contribute to its reduction through the legalisation of economic activity.

The factor of government effectiveness showed a positive correlation with the level of the shadow economy in Germany, Hungary, and Poland. This result differs from most empirical studies, which usually find that higher government effectiveness is associated with lower levels of informal economic activity (Buehn, Schneider, 2012; Schneider, Enste, 2013). It may indicate that strengthening the role of the state and improving the quality of administration do not always reduce informal economic activity. On the contrary, more effective public administration is often accompanied by more complex regulation, higher administrative requirements, and increased compliance costs, which may encourage workers and businesses to partially circumvent formal rules (Schneider, Enste, 2013). In such conditions, the shadow economy does not disappear but rather adapts and becomes less visible.

Personal income tax confirmed the hypothesis of its classic direct impact on the shadow economy and the motivation of workers to hide part of their income. A statistically significant effect of this factor was identified in the Czech Republic, Hungary and Slovakia, which may be due to a similar labour taxation structure and high

sensitivity of households to changes in net income. In these countries, personal income tax is perceived as a direct reduction in disposable income, and opportunities for partial underreporting of income remain relatively accessible, which strengthens incentives to participate in the informal economy (Schneider, Enste, 2002).

Gross domestic product per capita also exhibited a positive relationship with the level of the shadow economy, which is consistent with classical findings that link income levels to the scale of informal economic activity (La Porta, Shleifer, 2008). However, its effect proved to be considerably weaker compared to other factors included in the analysis. This may be explained by the scale of the indicator, whereby even substantial changes in income per capita are reflected in the dynamics of the shadow economy to a lesser extent. A statistically significant effect was identified in Germany, Hungary, and Slovakia, indicating that the shadow economy is sensitive to changes in overall income levels, while simultaneously confirming the secondary role of this factor relative to tax-related and institutional determinants.

A similar, though modest, positive effect was also identified for migration in Germany and Hungary. As with GDP per capita, the influence of migration proved to be weaker than that of tax-related and institutional variables. Migration may be linked to the shadow economy through difficulties in migrant integration and limited access to formal employment channels, including administrative and legal barriers, which can increase the likelihood of informal labour market participation (Zaiceva, 2014). In the context of intensified migration processes during the analysed period, migration thus appears as an additional, though secondary, factor influencing the scale of the shadow economy in selected countries.

Unemployment was found to be statistically significant in Germany and Hungary. This finding differs from the general pattern reported in the literature, which typically links higher unemployment to an expansion of informal economic activity due to reduced access to formal employment (World Bank, 2010; Elgin, Oyvatt, 2019). However, the direction of its effect differed between the two countries, highlighting the role of labour market institutions. In Germany, unemployment exhibits a conventional effect, whereby an increase in unemployment is associated with an expansion of the shadow economy. In contrast, in Hungary, higher unemployment is linked to a reduction in shadow economic activity, which may be explained by the sectoral structure of informal employment, concentrated in demand-driven and labour-intensive activities that are particularly sensitive to economic downturns.

Corporate income tax showed a statistically significant relationship with the level of the shadow economy only in Slovakia. In the remaining countries, the impact of this factor appears to be weaker and is likely outweighed by other determinants, such as the overall structure of taxation, business regulation, and general conditions in the corporate sector, which makes its independent effect harder to identify within the applied model. This pattern reflects mixed evidence in the literature: while corporate taxation is often linked to stronger incentives for informality (Djankov et al., 2010), some studies also show that the relationship is not universal and can be weak or even differ in sign depending on the institutional context (Giedraitis et al., 2023).

Taken together, the results indicate that the influence of individual determinants of the shadow economy in Central European countries is heterogeneous and largely shaped by national economic and institutional characteristics. Tax burden parameters and elements of the social protection system emerge as the most robust determinants, while macroeconomic indicators and migration play a supportive role. These findings underscore the importance of a comprehensive approach to the analysis of the shadow economy that accounts not only for the levels of individual variables but also for the way they operate within specific national contexts.

7.3 Limitations of the Study

One of the key limitations of this study is the limited availability of comparable statistical data across all countries and the entire period under analysis. For several potentially relevant indicators, particularly measures of money outside the banking system, no single harmonised data source exists. Data on the M0 monetary aggregate, which reflects the volume of cash in circulation, are published by national central banks in different formats and national currencies, significantly constraining cross-country comparability. Incorporating these indicators into the model would have required extensive data collection and harmonisation procedures, and could have reduced the homogeneity of the sample. Consequently, the analysis was restricted to the use of available and comparable macroeconomic data.

A further limitation concerns the treatment of the statutory retirement age in the model. In this study, retirement age is considered in an aggregated form and does not capture institutional differences across national pension systems, including gender-specific rules, child-related benefits, work-history-based entitlements, or early retirement schemes for specific occupational groups. This simplification was necessary to ensure cross-country data comparability; however, it may have led to an attenuation of the true impact of pension-related factors on the size of the shadow economy. Future research could therefore benefit from a more detailed comparative analysis of pension system design and its influence on incentives for formal and informal labour market participation.

From a methodological perspective, the study is based on the MIMIC model, which allows for a degree of flexibility in adapting the specification to available data, as implemented in this research. Nevertheless, the results remain sensitive to the chosen model specification, the set of explanatory variables, and the selected indicators. In addition, the representativeness of the findings is constrained by the focus on Central European countries and the selected time horizon, which limits the generalisability of the results to other regions or institutional settings.

8 Conclusion

The results of this study indicate that retirement age is not a key factor determining the size of the shadow economy in Central European countries. In most of the analysed countries, no statistically significant relationship was identified between changes in retirement age and the level of informal economic activity, suggesting that the impact of pension parameters depends on the institutional design of national pension systems. As a result, pension reforms alone are unlikely to affect the shadow economy without complementary institutional measures.

The analysis also revealed substantial cross-country differences in both the level and structure of the shadow economy. Despite similar historical and economic backgrounds, Central European countries differ in the stability of the shadow sector and in the factors shaping its dynamics. This confirms that the shadow economy is driven primarily by country-specific economic and regulatory conditions rather than uniform regional patterns.

Tax-related and institutional factors play a central role in explaining the dynamics of the shadow economy. In most countries in the sample, both social security contributions and self-employment exhibit an inverse relationship with its size. Higher social security contributions are associated with stronger compliance with formal employment rules and reporting obligations, while stable contribution frameworks may reduce the attractiveness of informal work by increasing the relative costs of operating outside the formal sector.

For self-employment, the inverse effect is mainly observed in countries with a well-developed institutional environment, where self-employment is largely formalised and integrated into the official economy rather than used as a channel for informal activity.

Other determinants generally demonstrate a positive relationship with the shadow economy, although their effects are less stable and vary across countries. Macroeconomic conditions, labour market characteristics, migration processes, government effectiveness, and the tax burden on labour and business influence informal economic activity depending on national conditions, labour market structures, and governance frameworks. In the Central European context, these results highlight the importance of labour market institutions and social protection systems shaped by post-transition reforms in determining responses to tax and regulatory incentives.

Overall, the findings emphasise that effective policies aimed at reducing the size of the shadow economy require a comprehensive approach focused on institutional quality and incentive structures within the formal sector. Uniform measures targeting individual parameters, including retirement age, are insufficient without considering national economic and institutional specificities.

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